

Mohammad Haider

Robotics Software Engineer

mhaider.biz@gmail.com | mhaider.dev | github.com/haidmoham | linkedin.com/in/haidmoham

SUMMARY

Software engineer with 4 years of experience building correctness-sensitive production systems, telemetry, and analytics infrastructure. Now applying that production discipline to robotics simulation in MuJoCo through measurable, reproducible experiments in controls, kinematics, and support mechanics.

EXPERIENCE

University of Virginia | Data Analyst

May 2025 - Mar 2026

- Owned Tableau analytics products for institutional stakeholders from discovery through implementation, deployment, and ongoing support.
- Translated ambiguous operational questions into SQL-backed metric definitions and decision-support dashboards.
- Collaborated with colleagues to investigate Tableau performance bottlenecks using telemetry, informing optimization work that improved organization-wide dashboard load times by approximately 20%.

Microsoft | Software Engineer - Data & Analytics

Jul 2021 - Jul 2024

- Migrated Microsoft's internal financial reporting platform from SSAS Multidimensional to Tabular with zero downtime, owning implementation, telemetry, production validation, and operational support.
- Extended automated financial-value reconciliation between legacy and replacement systems, preserving reporting correctness throughout the migration and improving data quality.
- Reduced representative filtering latency by up to 200x and user-visible Excel loading time by approximately 4x, measured through production telemetry and user reports.
- Built and maintained Azure-based financial reporting pipelines supporting business-critical planning and financial close operations.

Microsoft | Software Engineering Intern

Summers 2017, 2018

- Built internal engineering tools supporting reporting infrastructure and developer workflows across multiple engineering teams.

SELECTED PROJECTS

Robotics Test Bench + C-1N | Python, MuJoCo

- Built a MuJoCo test bench that turns robotics questions into prediction-led, single-variable experiments with deterministic headless runs, shared telemetry, and visual control/treatment overlays.
- Documented controller behavior, coupling, model mismatch, pose-dependent Jacobians, and static support through repeatable treatments, shared telemetry, and explicit stop boundaries; preserved each setup and failure for later comparison.
- Uses C-1N, a 12-actuator hexapod, as the integration target; implemented deterministic resets, interactive and headless rollouts, torso-orientation and foot-contact telemetry, and a preserved non-sustained SHUFFLE baseline.

Indigo Circuit | Flask, DuckDB, dbt

- Built and operated a Railway-hosted analytics platform with nightly idempotent ingestion and fail-closed shadow-database validation before atomic production swaps.

TECHNICAL SKILLS

Languages: Python, C#, SQL, Java, R

Robotics and simulation: MuJoCo, model-based control, inverse dynamics, Jacobians, task-space kinematics

Data platforms: DuckDB, dbt, Azure Data Factory, Azure Data Lake, SSAS Tabular

Engineering: Flask, PostgreSQL, REST APIs, Git, Azure, monitoring, validation, performance optimization

EDUCATION

Virginia Tech | B.S. Statistics